

SETS — ADVANCED PRACTICE PAPER

ICSE Mathematics · Class VII · Chapter: Sets

Total Marks: 80

Time Allowed: 2 Hours

General Instructions:

- All questions are compulsory. Marks for each section are shown in brackets.
- Show all rough work for Sections D and F in the space provided.
- For Section E, shade your answers neatly using a pencil.
- Write all sets and elements clearly, using proper set notation.

Section A — Multiple Choice Questions

(1 × 10 = 10)

A1. Which of the following is a well-defined set?

- (a) The collection of all rich people in a village
- (b) The collection of all multiples of 7 less than 50
- (c) The collection of the best cricket players in India
- (d) The collection of interesting books in a library

A2. The set $\{x \mid x \in \mathbb{N}, x^2 < 50\}$ in roster form is:

- (a) $\{1, 2, 3, 4, 5, 6, 7\}$
- (b) $\{1, 2, 3, 4, 5, 6, 7, 8\}$
- (c) $\{0, 1, 2, 3, 4, 5, 6, 7\}$
- (d) $\{1, 2, 3, 4, 5, 6\}$

A3. Which of these sets is infinite?

- (a) $\{x : x \text{ is a factor of } 100\}$
- (b) $\{x : x \in \mathbb{I}, x \text{ is a multiple of } 5\}$
- (c) $\{x : x \text{ is a day of the week}\}$
- (d) $\{x : x \text{ is a digit in the numeral } 2025\}$

A4. If $A = \{\text{letters of the word "ASSESSMENT"}\}$, then $n(A) = ?$

- (a) 10
- (b) 6
- (c) 5
- (d) 7

A5. Two sets A and B are called equivalent if:

- (a) $A = B$
- (b) $n(A) = n(B)$
- (c) $A \subset B$
- (d) $A \cap B = \varnothing$

A6. If $A = \{10, 20, 30, 40\}$ and $B = \{15, 25, 35, 45\}$, then $A \cap B = ?$

- (a) $\{10, 15, 20, 25, 30, 35, 40, 45\}$
- (b) $\varnothing$
- (c) A
- (d) B

A7. The number of subsets of a set with 5 elements is:

- (a) 5
- (b) 10
- (c) 25
- (d) 32

A8. If $U = \{1, 2, \dots, 12\}$ and $A = \{2, 3, 5, 7, 11\}$ (the primes ≤ 12), then $A' = ?$

- (a) $\{1, 4, 6, 8, 9, 10, 12\}$
- (b) $\{2, 3, 5, 7, 11\}$
- (c) $\{1, 2, \dots, 12\}$
- (d) $\varnothing$

A9. Which pair of sets below is equal?

- (a) $\{7, 8, 9\}$ and $\{9, 8, 7, 7\}$
- (b) $\{a, b, c\}$ and $\{1, 2, 3\}$
- (c) $\{x : x \in \mathbb{N}, x < 4\}$ and $\{1, 2, 3, 4\}$
- (d) $\{3, 6, 9\}$ and $\{3, 6, 9, 12\}$

A10. If $A = \{\text{factors of } 20\}$ and $B = \{\text{factors of } 30\}$, then $A \cap B = ?$

(a) $\{1, 2, 5, 10\}$

(b) $\{1, 2, 3, 4, 5, 6, 10, 15, 20, 30\}$

(c) $\{4, 20\}$

(d) $\varnothing$

Section B — Fill in the Blanks

(1 × 8 = 8)

B1. The empty set is denoted by the symbol _____ or _____.

B2. If $A = \{x \mid x \text{ is a vowel in the English alphabet}\}$, then $n(A) = \underline{\hspace{2cm}}$.

B3. Two sets having exactly the same elements are called _____ sets.

B4. A set having only one element is called a _____ set.

B5. If $A \subset B$ and $B \subset A$, then $A \underline{\hspace{2cm}} B$.

B6. The symbol used to denote "belongs to" is _____.

B7. If $n(A) = 4$, the number of subsets of A is _____.

B8. $A \cap B = \varnothing$ means A and B are _____ sets.

Section C — True / False (with justification)

(2 × 6 = 12)

C1. $\{\varnothing\}$ is an empty set.

☐ True

☐ False

C2. If $A \subset B$ and $B \subset C$, then $A \subset C$.

☐ True

☐ False

C3. If $A = \{4, 6, 8\}$ and $B = \{8, 6, 4\}$, then $A \neq B$.

☐ True

☐ False

C4. If A and B are disjoint sets, then $A - B = A$.

☐ True

☐ False

C5. The cardinal number of an infinite set can be found by counting all its elements one by one.

☐ True

☐ False

C6. If $A \cap B = A$, then $A \subset B$.

☐ True

☐ False

Section D — Short Answer Questions**(3 × 6 = 18)**

D1. Write the set $A = \{x \mid x \in \mathbb{N}, x \text{ is a multiple of } 3, x < 20\}$ in roster form.

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D2. Write the set $B = \{7, 14, 21, 28, 35\}$ in set-builder form.

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D3. If $P = \{\text{letters of the word "MISSISSIPPI"}\}$, find $n(P)$. Show the distinct letters you counted.

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D4. For each pair below, state whether the sets are **equal**, **equivalent only**, or **neither**. Justify each briefly.

(i) $A = \{\text{Monday, Tuesday, Wednesday}\}$, $B = \{12, 24, 36\}$ (ii) $A = \{7, 9, 11\}$, $B = \{11, 9, 7\}$ (iii) $A = \{9, 18, 27\}$, $B = \{9, 18, 27, 36, 45, 54\}$

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D5. List all the subsets of the set $A = \{x, y, z\}$. How many subsets are there in total?

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D6. If $U = \{1, 2, 3, \dots, 15\}$, $A = \{\text{multiples of } 3 \text{ in } U\}$, and $B = \{\text{multiples of } 5 \text{ in } U\}$, find:

(i) $A \cup B$ (ii) $A \cap B$ (iii) $A - B$ (iv) B'

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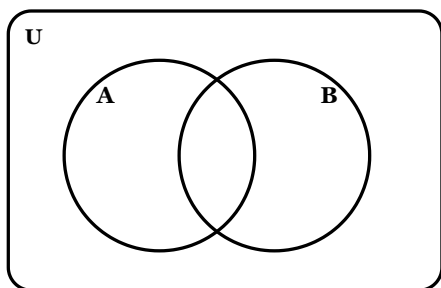
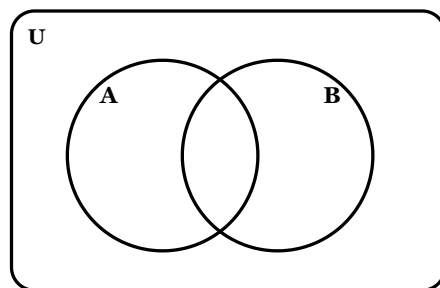
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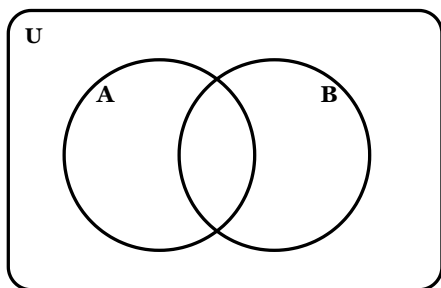
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Section E — Venn Diagrams (draw and shade)**(3 × 4 = 12)**

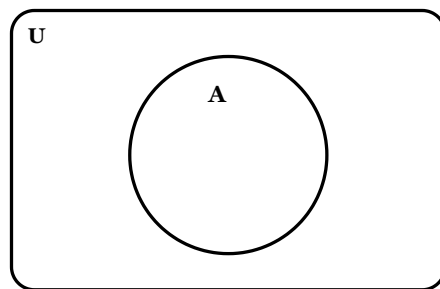
For questions E1–E3, use $A = \{2, 3, 4, 5, 6\}$ and $B = \{5, 6, 7, 8\}$. Shade the region of each diagram that represents the given expression.

E1. Shade $A \cup B$ **E2. Shade $A \cap B$** 

E3. Shade $A - B$



E4. Let $U = \{1, 2, \dots, 9\}$, $A = \{3, 6, 9\}$. Shade A'



F1. $A = \{\text{students in the reading club}\} = \{\text{Priya, Rohan, Sanya, Vikram}\}.$

$B = \{\text{students in the dance club}\} = \{\text{Sanya, Vikram, Tanvi}\}.$

A new student, **Meher**, joins the school and joins **both** the reading club and the dance club.

Rewrite the sets A , B , $A \cup B$, and $A \cap B$ **after** Meher joins.

F2. Let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $B = \{4, 5, 6, 7, 8, 9, 10\}.$

(a) Find $n(A)$, $n(B)$, $n(A \cap B)$, and $n(A \cup B)$ by direct counting.

(b) Using your answers, check whether $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ holds true for this example. Show your working.

F3. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 2, 3, 6\}$, and $B = \{2, 4, 5, 7\}.$

(a) Find A' and B' .

(b) Find $A \cup B$, and hence find $(A \cup B)'.$

(c) Find $A' \cap B'$, and verify that $(A \cup B)' = A' \cap B'$ (this is called De Morgan's Law).

F4. Let $A = \{4, 8, 12, 16\}$, $B = \{12, 16, 20, 24\}$, and $C = \{20, 24, 28, 32\}.$

Find: (i) $A \cup B$ (ii) $B \cup C$ (iii) $A \cup B \cup C.$
